

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-7; AGRICULTURE-4

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2. AGRI-MARKETING REFORMS

1) SYLLABUS:

- Transport and marketing of agricultural produce and issues and related constraints

2) AGRI-MARKET REFORMS

- Introduction

- » NITI Aayog has identified '**Agriculture Market Reforms**' as one of the three key areas of reform in Agriculture sector.
- » **Efficient Agri-marketing is crucial for:**
 1. **Ensuring proper prices for farmers** -> promotes and sustains agricultural production and productivity -> food security and inclusive growth
 2. **Lower prices for consumers** -> efficient marketing mechanisms reduce the cost of transaction and share of intermediaries.
 3. **Reduces post-harvest losses** -> by providing quick transportation and storage facilities
 4. **Stimulating agro-based and food processing industrial growth:** Ensures affordable availability of raw material for agro-based industries.
 5. **Reduced subsidy burden for government** -> Efficient marketing will reduce the need of procurement by government from MSP.
- » However, the Agri development in India has ignored the potential of marketing and has continued to follow its old trajectory. Since, agri-marketing is a state subject, it is stuck in a tussle between Centre and states.

- How has the system functioned so far? -> The APMC System and its issues

- » The Agri markets in India are mainly regulated by the state Agriculture Produce Marketing Committee (APMC) laws. These laws enable state governments to regulated wholesale markets and marketing practices. APMC Acts make it mandatory for farmers to sell their produce only to licensed merchants (can also be called government approved merchants) at Mandis set up by State Agriculture Market Boards.
- » **What was the need of APMC Laws / Regulated Markets (Mandis)**
 - In the absence of any market regulations, the farmers were being exploited by traders and intermediaries.
 - The regulated Mandis prevented the exploitation by ensuring **fair prices through bidding process** at these regulated markets.
- » **Current Situation**
 - India currently has around 2500 principle regulated primary agricultural markets in the country which are governed by state APMC Acts and administered by a separate Agricultural Produce Marketing Committee.
 - APMCs regulate the trade of farmers by
 - Providing licenses to buyers, commission agents, and private markets.
 - Levy market fees or any other charges on such trade
 - Provide necessary infrastructure within their market to facilitate trade.

- Trade is allowed withing the APMC mandis or between two APMC mandis situated in the same state.
- » **How these Mandis have served farmers:** These APMCs have ensured that fruits of Green Revolution reach farmers.
 - **Reduction in exploitation by intermediaries** - not allowed to sell products outside the mandi
 - **Good Price: Sale through auction** -> best possible prices available.
- **But, present APMC structure has led to many challenges/limitations**
 - » **APMC Mandis are inaccessible for most of the farmers:** The average areas served by mandis in the country is 434.48 sq km, whereas the National Commission on Farmers (2006) have recommended market within a radius of 5 km (corresponding to market area of about 80 sq km). This makes it very difficult for small and marginal farmers to reach these mandis.
 - » **Price Fixing, Cartelization** by licensed traders -> Traders, commission agents, and other functionaries organize themselves into associations, which **don't allow easy entry of new persons into market yards, stifling competition**.
 - » **Information asymmetry** and opaque process for price discovery.
 - » **Undue deduction** in the form of commission charges and market fees.
 - » **Fragmentation of agri-market**
 - Hinders free flow of agri-commodity, not only between states, but within states too
 - Escalate prices because of multiple handling
 - Difficulty in very large procurement by food processors or exporters
 - » **Hindered private investment in agri-marketing -> Poor Infrastructure and low use of technology**
 - As according to APMC acts, only state governments could set up these Mandis.
- **To remove the above limitations, the central government has taken many steps:**
 1. **Model APMC Acts** (in 2003 and 2017-18), have given suggestions for liberalization of agri-markets.
 2. Promotion of **e-NAM** by central government for promotion of unification of markets both at state and national level.
 3. **Bihar** has completely done away with APMC act and **Kerala** had never implemented it. **Maharashtra** is also trying to liberalize the state APMC law.
 4. Development primary markets / submarkets as **GrAMs** (Gramin Agriculture Markets) with improved infrastructure
 5. Creation of **Agri Market Infrastructure Fund (AMIF)** for development and upgradation of Gramin Agriculture Markets.
 6. The three farm laws of 2020 was also an attempt to reform the agri-marketing system in India. But these laws had to be repealed due to huge farmer protest.
- **Way Forward**
 - » **Reform APMCS** (responsibility of states) -> Incentivize states to adopt and implement features of model laws.
 - **Ease entry of private players**
 - **Reduce trader collusion**

- **Link them with national e-marketing portal.** The introduction of unified national licenses for traders and a single point of levy of market fees are also steps in the right direction.
- **Delink the regulatory and operational role** of APMCs.
- » **Empower farmers to given them more bargaining powers: FPOs** can play a crucial role in this.
- » **Strengthen Agri-Future Markets** which is a standard way to of hedging risks in market economy.
- » **Deal with other factors of agrarian crisis:** Monsoon dependent agriculture, climatic vagaries, lack of access to technology that can boost productivity; and lack of institutional finance and affordable rates.
- » Develop a **Price Stabilization Scheme** mechanism to give a lift to market prices whenever market price crash.
- » **Centre should focus on providing stable and predictable policy environments** viz imports-exports; functioning of national commodity exchanges and future markets, and providing **inclusive platforms for discussions on** state-level market reforms, public procurement and price support, designating safeguard against consolidation of corporate interests and framing data policies.
- » **Learn lessons from 3 farm laws saga:** Process of economic reforms must be more consultative, more transparent, and better communicated to the potential beneficiaries.

3) FUTURE MARKET FOR AGRI-COMMODITIES (FROM ESI 2023-24)

- **Introduction:** The future market serves as a mechanism that brings the prospects of future production and consumption to influence today's price in a logical manner. It leads to inter-temporal smoothing of prices.
- **Evolution of Commodity Derivative market in India:**
 - » The commodity derivative market has a long history in India. During 1940s, India had around 300 commodity exchanges. But till 1952, trading on these exchanges was not regulated by a market regulator.
 - » In 1952, GoI formulated the Forward Contracts (regulation) Act of 1952 and set up Forward Market Commission (FMC) in 1953 as the regulator.
 - » In 1966, a comprehensive ban on futures trading was imposed to control price volatility.
 - » The **revival of agriculture future trade in India** happened in 2003 with the setting up of NCDEX, MCX and NCME. In 2015, SEBI took over as the regulator of the commodity market.
 - » In parallel, e-NAM was introduced as electronic online marketplace for agri-produce.
- **Significance of commodity future market:**
 - » **Better price discovery** and thus better prize realization for farmers.
 - » **Reduces temporal variability** in prices.
- **Key factors hindering futures market of Agri-commodities:**
 - » **Very few participants:** Commodity future market can ensure better price discovery only when many consumers, producers, traders and aggregators used these markets to hedge their risks.
 - » **Most Indian farmers are small and marginal:** They don't have necessary wherewithal to engage/participate in these markets effectively.

- » **Wide variety of commodities in different qualities** are produced in India because of some much diversity in geography, seeds, soil etc. But the commodity market require standardized exchange contracts with specified quality parameters and delivery requirements.
 - » **Policy inconsistency**: For e.g. the periodic bans imposed by GOI on futures trading on agri-commodities as one of the measures to counter food inflation.
- **Way Forward:**
- » **Abhijit Sen Committee** have recommended to identify those commodities where there is possibility of future trading affecting expectations that may influence inflation in essential commodities and insulate these from futures. The agriculture future market may focus on less sensitive commodities like oilseed complex (oilseeds, meals, and oils), Feed (maize), cotton, basmati rice, spices etc.
 - » **Enabling small and marginal farmers**: NABARD is responsible for forming 10,000 FPOs. It should also device a compulsory module that train FPOs to navigate the realm of agri-futures.
 - » **Government Agencies** - Food Corporation of India, NAFED etc should also increase their participation in agri-futures.
 - » **Certainty in government policies** - unpredictable and restrictive policies (like banning of agri-exports or futures) leads to problems.
- **Conclusion**: Along with physical agri-markets, future markets are crucial for increasing farm income and therefore, the government should reduce the impediments to the success of future market in agriculture sector.

3. E-TECHNOLOGY IN AID OF FARMERS

1) E-TECHNOLOGY IN AID OF FARMERS

- **Question:**
 - » Discuss the concept of digital transformation in agriculture and its significance in enhancing agricultural productivity, sustainability, and farmers' livelihoods in India [10 marks, 150 words]
 - » Discuss some recent initiative focused on promoting e-technology in agriculture. What are the challenges hindering the digital transformation of agriculture sector in India [15 marks, 250 words]
- **Introduction:**
 - » Digital technologies are bringing about a revolution in almost all aspects of our lives today. India's agriculture sector is also undergoing a significant transformation with the integration of digital technologies.
- **Key Government Efforts:**
 - » **The Digital Agriculture Mission 2021-2025** aims to modernize agriculture through advanced technologies like AI, remote sensing, drones etc.
 - » In Budget 2023-24, government has also announced **Digital Public Infrastructure (DPI)** for agriculture as an open source, open standard, interoperable public good. It will enable inclusive, farmer-centric solutions through relevant information services for crop planning and health,

improved access to farm inputs, credit, and insurance, help for crop estimation, market intelligence, and support for growth of agri- tech industry and start-ups.

- **Agri-Stack** is one of the prominent DPIs with three foundational registries (databases) i.e. **Farmer's Registry/Database**, **Geo-Referenced village maps** and the **Crop sown registry/database**, along with several support registry database.
 - The three foundational registries will enable **digitally authenticable identities and non-repudiable digital assets** for the farmer in the form of **Farmer IDs**, **geo-tagged farm plots**, and **Crop sown data**.
 - **The Krishi Decision Support System (Krishi-DSS)** is another DPI, which aims to **integrate and store in a standardized form relevant geospatial and non-geospatial data**, such as remote sensing data, weather data, soil data, crop signature library, reservoir data, groundwater data, and data pertaining to government schemes.
 - » **Krishi Mapper** - a geospatial **mobile application** for all the **land based schemes**, which enables **geo-fencing** (polygon creation/ latitude longitude) capture also incorporating **Geotagged photographs** from the current location survey/ inspection
 - » **Comprehensive Soil Fertility and Profile Mapping** - for suitable soil-health related interventions
 - » **Digital General Crop Estimation Survey** - accurately measure crop yields, **through crop cutting experiments** on randomly selected plots.
 - » **Support to Agri-Tech Startups**: As of Feb 2024, **554 agri-start-ups**, including **387 women led startups**, are working in the agriculture & allied sector.
 - » **National Strategy for AI** recognizes **agriculture as one of the priority areas** for implementation of AI driven solution
- **Digital transformation of Agri-food system** may contribute in the following manners:
- » **Increasing efficiency on the farm**: Technologies like AI, ML, IOT can play a very important role in making resource **efficient precision agriculture** possible.
 - » **Choosing the right Crops**: With the help of modern **predictive analytics**, farmers can even be **more informed about the potential success of a particular crop**.
 - » **Improving farmer's access to input, output and financial markets**:
 - **FARMS-APP** - allows farmers to connect with the custom hiring centres for renting equipment.
 - For e.g. **E-NAM** has made possible for farmers to send her produce to traders anywhere in the country.
 - **PM-KISAN mobile app** broadens the **reach of the scheme** where farmers can view the **status of their application**, update or carry out corrections of name based on their Aadhaar card and also check the history of credits to their bank accounts.
 - » **Estimating Agri-Production**:
 - **The MNFC App**, developed by National Remote Sensing Centre, ISRO, uses satellites to collect field data for crops assessment under the FASAL project of the Ministry of Agriculture.
 - **Predicting Crop Yield**: Many companies are **developing models to predict yield** using satellite data, administrative data and weather data through AI-ML models.
 - » **Strengthening quality control and traceability**.
 - **Grading Farm Produce: Handheld Gadgets**, along with **image recognition powered by AI-ML** are being developed to **grade farm produce** another key constraint in agri-marketing and agri-exports.

- **Ensuring Traceability:** Several pilots are underway in the country, where **blockchain platforms are being developed** to provide end-to-end traceability of agriculture produce, a key constraint in growing our export base. E.g. -> Basmati.net, HortiNet Mango, TraceNet, MeatNet etc.
 - » **Market Intelligence:** APEDA and MPEDA are running 'agri-exchange' and 'fish exchange' portals, respectively to provide market intelligence to their stakeholders.
 - » Improving the design and delivery of agriculture policies.
- **Factors which is leading to digitization of Agriculture area:**
 - » Almost complete penetration of smart phones in rural India.
 - » Many private players are looking at the digital agriculture opportunities in India and are taking a lot of steps. Multinational companies like Microsoft, Amazon and IBM are working towards improving the situation.
 - E.g. MoA&FW have signed MoU with Microsoft to run a pilot program for 100 villages in 6 states. Here Microsoft will create a 'Unified Farmer Services Interface' through its cloud computing services
- **Challenges and Constraints**
 - » Poor digital literacy; lack of awareness;
 - » Increasing cases of digital frauds are making farmers wary of using e-technology.
 - » Language barriers and less content localization: Most digital tools and application in agriculture are available in English, whereas most of the farmers understand regional languages.
 - » Infrastructure challenges like lack of electricity, storage facilities etc.
 - » Data Privacy and Cyber Security
 - » Small and marginal landholdings
 - » High initial cost of equipment for precision agriculture etc. makes it difficult for small and marginal farmers to buy it.
- **Conclusion:** Addressing these challenges requires concerted efforts from the government, technology providers, financial institutions, and agricultural stakeholders. Infrastructure development, digital literacy programs, localization of content, and policy frameworks that safeguard data privacy and promote inclusive access are vital to accelerate the digital transformation of the agriculture sector in India

2) E-NAM

- **Introduction: e-NAM**
 - » e-NAM is a pan India **electronic trading portal** for farm produce which creates a **unified national market** for agricultural commodities by integrating existing APMC markets and other market yards. It was launched in **April 2016.**
 - » This portal provides a **single window service** for all APMC related services and information, such as commodity arrivals and prices, buy and sell trade offers etc.
 - » Under e-NAM, the Government provides **free software and one time assistance of Rs 75 lakh per mandi** for computer hardware and IT infrastructure.
- **Salient Features**

- **National e-market platform** for transparent sale transaction.
 - **Enables farmers to showcase their produce** through their nearby markets and **facilitate traders from anywhere to quote price.**
 - **Liberal licensing** of traders / buyers and commission agents by state authorities.
 - There are **no preconditions for physical presence** or possession of shop/premises in the market yard.
 - **One license** for the trader would be valid all across the country.
 - **Single point levy of the market fee**, i.e. on the first wholesale purchase from the farmer
 - **Harmonizing the quality standards of agricultural produce** and infrastructure for quality testing is made available in every market to enable **informed bidding** by buyers. At present, **Common tradeable parameters** have been developed in 25 commodities.
 - **States desirous to join** has to accordingly **enact suitable provisions in their APMC Act.**
 - The amendments include a **single point levy of mandi fee**, **unified trade license valid across all mandis of the state**, and provision for **e-auction facilities**.
 - **States can have their own electronic platform** and can **decide to link them to NAM.**
- **Progress as of April 2024:**
- **1.7 crore farmers** and **2.56 lakh traders** have been registered on the e-NAM portal; **8195 FPOs** have also registered under the new FPO schemes.
 - **Trading parameters** for around 200 agricultural and horticultural crops have been provided.
 - **Warehouse based trading modules** are provided in e-NAM to facilitate trade from warehouse based on e-NWR
- **Advantages**
- **Farmers**
 - **Open price discovery** and **better returns to farmers**: As per ESI 2023-24, e-NAM has had a positive impact on farmers and have **enabled higher prices for farmers**. Farmers have also found **quality testing procedure transparent**.
 - Farmers **get more options** for selling their produce.
 - It has also **reduced transaction cost**.
 - **For Buyers/ Traders**
 - **Source commodity from any mandi.**
 - Physical presence and dependence on intermediaries not required, thereby **reducing the intermediation cost**
 - Local traders can **get large national market** for secondary trading.
 - **Bulk buyers** have specifically benefitted as they are now **able to participate and procure directly** at the local Mandi level through e-NAM platform, thereby reducing their intermediation costs.
 - **Consumers**
 - Increase number of traders and thus **competition** would increase. This would lead to **lower and stable prices**.
 - **Mandis**
 - There will be **reduction in book keeping and reporting system** as it will be generated automatically.

- Monitoring and regulation of traders and commission agents becomes easy.
 - Transparency in the process eliminates the scope of manipulation of tendering/auctioning process.
 - Market allocation fee will increase due to accounting of all transactions taking place in the market.
 - It will reduce the manpower requirements as the tendering/auctioning process is carried out electronically.
 - For instance, the system declares the winner of lots within few seconds.
 - It eliminates information asymmetry as all the of an APMC can be known directly from the website
- **But, the volume of trade of e-NAM and number of traders and farmers registering on e-NAM has been pretty low.** It is because e-NAM suffers from many limitations.
- » **Lack of awareness among farmers** regarding e-NAM and its usefulness.
 - » **Small and Marginal farmers find e-NAM unattractive**
 - Most farmers sell their yield to a local produce aggregator than taking their crops to Mandis.
 - Even if farmers take their produce to mandi their yield would be very small to attract a distant trader.
 - » **Non-Uniform Quality Standards across different states**
 - » **Infrastructure issues** related to setting up of assaying facilities.
 - » **Obligations on states for successful implementation** has limited the rate of progress of e-NAM.
- **Way forward**
- » **Train various stakeholders** and make them aware of the advantages of e-NAM.
 - » **Harmonization of quality standard** across the states is the need of the hour. SFAC is already exploring the opportunity to bring uniformity in quality standard. Government should be setting up quality labs in all the APMCs to harmonize the quality standards.
 - » **Small farmers can aggregate their produce** at their own bypassing local producer aggregator.
 - Cooperatives and farmer produce organizations can play a very important role
 - » Further, **fast track the integration of e-NAM with GrAMs**
 - » Government should try to route purchase of paddy under MSP mechanism through the e-NAM platform. This should be done at least in the states that have, on principle, allowed their mandis to link up to the portal.
 - » **Strengthen Storage and Transport facilities** to ensure that online bought products reach the buyer at affordable rate and in good conditions.
 - » **Expand the role of e-NAM:** The framework of e-NAM can be evolved from being a marketing portal to a one-stop information centre for accessing government policies, weather conditions, Early Warning Systems, and extension services, links to financial institutions for credit insurance etc.

4. FOOD PROCESSING INDUSTRIES (FPIS)

1) SYLLABUS:

- Food processing and related industries in India- scope and significance, location, upstream and downstream requirements, supply chain management

2) PYQS

- What are the impediments in marketing and supply chain management in industry in India? Can e-commerce help in overcoming these bottlenecks? [Mains 2015]
- What are the reasons for poor acceptance of cost-effective small processing unit? How the food processing unit will be helpful to uplift the socio-economic status of poor farmers? [Mains 2017] [15 marks, 250 words]
- Examine the role of supermarkets in supply chain management of fruits, vegetables and food items. How do they eliminate number of intermediaries? [Mains 2018] [15 marks, 250 words]
- Elaborate the Policy taken by Government of India to meet the challenges of the food processing sector. [Mains 2019] [15 marks, 250 words]
- What are the challenges and opportunities of food processing sector in the country? How can income of farmers be substantially increased by encouraging food processing [Mains 2020] [10 marks, 150 words]
- Elaborate the scope and significance of Food Processing Industry in India [Mains 2022, 10 marks, 150 words]

3) FOOD PROCESSING INDUSTRY (FPI)

- **Introduction**
 - » The Food Processing Sector has emerged as an **important segment of Indian economy** in terms of its contribution to GDP, employment and investment. It is a high growth and high profit sector due to its immense future potential for value addition. During the **last eight years ending 2022-23**, the food processing industries has been growing at an average annual growth rate (AAGR) of around **5.35% at 2011-12 prices**.
- **Current Status of Food Processing in India**
 - » India ranks fairly low in the global food processing value chain. In spite of large production base, **very small % of food items are processed in India** (Vegetables and fruits (2%), Marine items (8%), milk (35%) and poultry (6%))
- **Scope of Food Processing Industries**
 - » **Resource Advantage of India - Availability of Raw Material:** India is among the largest producers of fruits, vegetables, dairy and meat products. With this base, we can easily become the largest supplier of processed food in the world.
 - » **Increasing Consumer demand for processed food:** With changing socio-economic conditions in the country and increasing income of the working class, the demand for processed food is increasing.
 - » **Vast labor force** of India is also looking for employment opportunities.

- » **Export Potential** of this sector is huge as the demand for processed food is not only very high globally, but also expected to increase further with increasing population and decreasing agri-outputs with climate change.
 - Further, India is surrounded by food importing countries like Maldives, Sri Lanka, Myanmar etc.
- **Significance of Food Processing Sector**
 - » **Economic Growth:** The sector contributes to 7.66% of GVA of manufacturing sector in 2022-23 with a GVA of 1.92 lakh crore.
 - » **Employment:** It is an employment intensive sector and have 20.05 lakh employees with 12.2% of total persons engaged in registered manufacturing sector.
 - » **Potential to boost Exports:** At present India's agri exports primarily consist of raw materials, which are then processed in other countries. Therefore, we have a lot of space to move up in the value chain.
 - » **Reducing Wastage of food material,** especially the perishable food items.
 - Processed food can be stored for longer and thus also limits price fluctuation due to variability of agri-output.
 - According to UN India wastes around 30% of the food produced. Similarly, NITI Aayog estimates the annual harvest losses of close to Rs 90,000 crores.
 - » **Increasing farmer's income by adding value to produce** and reducing waste.
 - » **Promotes diversification of crops** as the Food processing industries require different types of crops.
 - » **Promoting industries in Rural Areas** and thus **employment** and entrepreneurial opportunities in rural areas. Moreover, it is an employment intensive sector and would lead to advantages like reducing rural-urban migration.
 - As per ESI 2023-24, the FPI in India is one of the largest employers in organized manufacturing, with a 12.02% share in the total employment in the organized sector.
 - » **Nutritional Security** is another area where Food Processing Industries Contribute. By fortifying processed food with micro-nutrients a number of nutritional challenges can be dealt with.
- **Upstream Requirements**
 - » Cultivation of required raw material
 - » Modern Extraction techniques
 - » Good Linkages of FPI with farmers
 - » Storage facilities for raw materials like grains, meat, fish etc.
 - » Quality testing Facilities
 - » Skilled Workforce
 - » Efficient transportation facilities
- **Downstream Requirements** deal with processing of material collected during upstream phase into final product.
 - » The key requirements of downstream side are:
 - a. Latest processing techniques
 - b. Latest processing machinery
 - c. Quality testing facilities
 - d. Efficient Transportation facilities to distribute the final product

e. Skilled workforce to handle all the above requirements

- **Supply Chain Management in Food Processing sector**

- » Supply chain management is the **management of the flow of goods and services** and includes all processes that transform raw material into final products and makes final product reach consumers.
- » It involves active streamlining of business' supply side activities (upstream requirements) and downstream activities.
- » The supply chain management is the management of procurement of raw material, production, shipment and distribution of the final product to ensure a competitive high quality product reaches the consumer.

- **Despite its huge potential, the food processing sector contributes way below its potential in the Indian Economy because of the following reasons:**

» **Industry level Issues**

- **Informal, Unorganized Food Processing Sector** faces a number of challenges which limit their performance and their growth.
 - Lack of **access to modern technology and equipment, training, access to institutional credit**, lack of basic awareness **on quality control** of products, and **lack of branding and marketing skills**.
- **Entry Barriers - Capital Intensive** nature of the sector becomes an entry barrier and prevents small players to enter the sector.
- **Small Size of Indian Companies** -> **less R&D and marketing** compared to Nestle, PepsiCo etc. -> **Difficult to compete**
- **Laboratories**: Lack of High-Quality Processed food testing labs -> prevents standardization and certifications -> hampers exports
- **Lack of skilled human resource** -> hampers growth of the sector

» **Problems at the level of Agriculture / Primary Production**

- **Poor Market Reforms** (for e.g. in APMCs) **in the Agri Sector** -> Hinder availability of raw material
- **Slow pace of growth in contract farming and land leasing reforms**
- **Government Policies Affecting market prices** [e.g. sudden ban on import/exports etc.] -> hampers ease of doing business
- **Preprocessing losses** occur due to lack of awareness and a dearth of storage and packaging facilities close to the farm gates.
- **Infrastructure Deficiencies** in transportation, storage etc also leads to lots of losses and make our food processing sector less competitive
- **Variation of Quality** is another impediment. Lack of avenues to adequately grade, sort and pack perishable produce is a major culprit in this regard.

- **Steps taken to promote FPI**

1. **Production Linked Incentive (PLI) Scheme for the Food Processing Industry (PLISFPI)**: It supports creation of global manufacturing champions, branding and marketing abroad.

- It is a central sector scheme for implementation during 2021-22 to 2026-27 with an outlay of Rs 10,900 crores. It is aimed at increasing the production capacity and improve international branding for the food processing sector in India.
- 2. **Pradhan Mantri Formalization of Micro Food Processing Enterprise (PM FME):** With an outlay of Rs 10,000 crores, the scheme provides credit linked subsidies to individuals/SHGs/FPOs etc. for food processing infra development. It also helps in development of common infrastructure like labs, warehouses etc.; support for marketing and branding; training; product development; packaging etc. it is based on **One District One Product (ODOP)** approach.
- 3. **Other Infrastructure Improvement**
 - **Pradhan Mantri Kisan SAMPADA Yojana** (Scheme for Agro-Marine Processing and Development of Agro-Processing Clusters) - a central sector scheme with an allocation of Rs 6,000 crore for the period 2016-20 for creation of modern infrastructure with efficient supply chain in the food processing sector.
 - It has **incorporated other schemes** related to food processing under it including Mega Food Parks, Integrated Cold Chain and Value Addition Infrastructure, Creation/Expansion of Food Processing/Preservation Capacities, Infrastructure for Agro-Processing Clusters, Creation of Backward and Forward Linkages, Food Safety and Quality Assurance Infrastructure, and Human Resource and Institutions.
 - **Mega Food Park Scheme, 2008:** It focuses on establishing mega food clusters for creating major infrastructure facilities in India to add value and reduce wastage at each stage of the supply chain (from farm to market). MoFPI provides a assistance of 50% of the project cost (excluding land), subject to a maximum of Rs 50 crore.
 - **Modernization of Abattoirs Scheme:** Enhance processing and preservation capacities to improve quality and reduce wastage.
- 4. **FPI is One of the priority sector under Make in India initiatives**
- 5. **Steps to ensure Credit Availability**
 - a. The government has set up a Special Fund of Rs 2,000 crore in NABARD to make available affordable credit at concessional rate of interest to designated food parks and agro-processing units.
 - b. Food and Agro Processing Units and Cold Chain infra have been brought under the ambit of **Priority Sector Lending**
 - c. **Subsidized credit through PM FME**
- 6. **Strengthening of Food testing Infrastructure** in India by upgradation of FSSAI labs and setting up of new labs in the country.
- 7. **Scheme for Human Resources and Institution - Skill Development (SHRISD)**
 - To ensure supply of high quality human resource in the food processing sector. The scheme focuses on skilling infrastructure and development of course curriculum.
- 8. **Operation Green:** Ensuring subsidized transportation and storage for 22 notified perishable fruits and vegetables. The scheme's objectives include enhancing farmers' value realisation, reducing post-harvest losses, increasing food processing capacities, and adding value. The scheme has two-pronged strategies: Price Stabilization Measures (short-term measures) and Integrated Value Chain Development Projects (long-term).

9. **100% FDI** is permitted under the automatic route for food processing sector.
- **For e-commerce** in respect of food products manufactured and/or produced in India, 100% FDI through approval route is allowed.

- **More steps that can be taken:**

- » Access of MSME sector to Finance is a perennial problem. **Smart financing alternatives** such as peer-to-peer (P2P) lending hold potential for micro food processors.
- » **Strengthening TReDS:** Access to working capital has in theory been addressed by the Trade Receivables Discounting System (TReDS), a platform for facilitating the financing/discounting of trade receivables of MSMEs through multiple financiers.
 - However, the platform requires considerable scaling-up and simultaneous enforcement of stringent measures for corporates to comply with.
 - Integrating TReDS platform with GSTN's e-invoicing portal will make TReDS more attractive and give relief to financiers.
- » **Increasing Demand for processed food:** The industry in India should focus on new alternative types of food including nutri-cereals, plant-based proteins, fermented foods, health bars and even fresh fortified foods for pet.
- » **Improving Infrastructure:** Infrastructure bottlenecks can be dealt with through farmer-producer-investors-R&D lab linkages.
- » **Effective Implementation of** existing schemes programs - PMKSY, PM FME, PLIS etc.
- » **Streamline** the general challenges like labor law reforms, regulatory structure etc.

- **Conclusion**

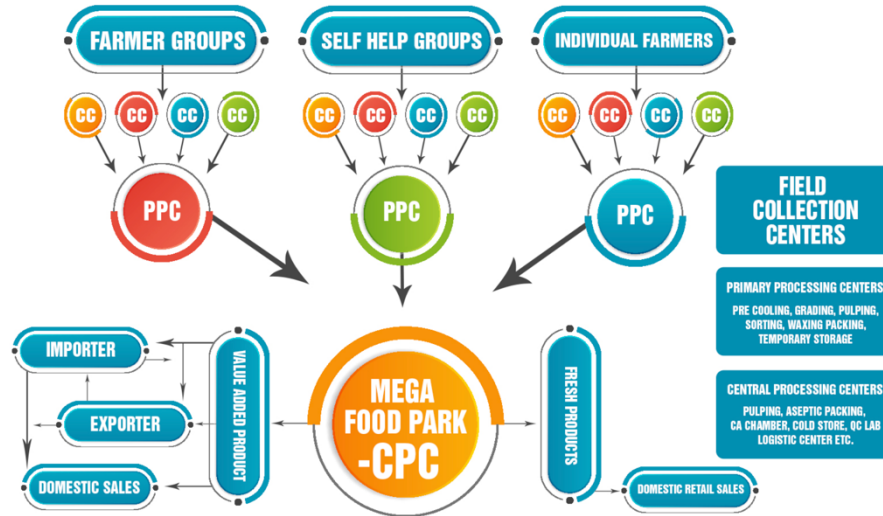
- » Food Processing Industries have the potential to solve some of the key challenges faced by India - promoting exports, increasing farmers income, rural industrialization, reduce rural urban migration, reducing food wastage etc.
- » Keeping this in mind, government has taken various initiatives to promote FPIs in India. Now, the focus should be on ensuring effective implementation of these schemes through proper investment, clear focus and strong political will.

4) MEGA FOOD PARK SCHEME: ANALYSIS

- **Ministry:** MoFPI
- This scheme is now a component of Pradhan Mantri Kisan Sampada Yojana (PMKSY).
- It focuses on establishing mega food clusters for creating major infrastructure facilities in India to add value and reduce wastage at each stage of the supply chain (from farm to market). MoFPI **provides a assistance of 50% [75% in NE and Hilly states] of the project cost** (excluding land), subject to a maximum of Rs 50 crore.
- The park is developed on **Hub and Spoke Model**, where Central Processing Centre acts as Hub and PPCs and CCs act as Spokes.
 - Facilities for **primary processing and storage** is created near the farm in the form of **Primary Processing Centre (PPC)** and **Collection Centre (CC)**.

- Common facilities and enabling infrastructure for providing **secondary and tertiary infrastructure and processing** services is created and **Central Processing Centre (CPC)**.

MEGA FOOD PARK MODEL



- Under the scheme government has sanctioned setting up of **42 Mega Food Parks** in the country.
 - The scheme has **been discontinued by government with effect from 1st April 2024** with provisions for committed liabilities for ongoing projects only.
- Advantages**
 - » Contributes to reduction of food wastage, increased income of farmers, better value addition etc.
 - » Gives boost to food processing sector (hence to export and employment opportunities) by creation of high-quality infrastructure.
 - » Capacity building of producers and processors and creation of efficient supply chain along with significant direct and indirect employment generations.
- Key challenges**
 - » Bureaucratic bottlenecks -> state governments taking a lot of time for statutory clearance
 - » Lack of quality human resource to work on the MFPs.
 - » Lack of awareness about the scheme.